Madda Walabu University

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Curriculum of Master of Science in Physics

by Department of Physics

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to be submitted to School of Post Graduate

Preface

The role of Physics in the economic development of a nation and the prosperity of its people is very important and it is imperative for a nation to train its workforce in contemporary Physics. At the same time, knowledge is expanding rapidly and new courses and disciplines are emerging at a fast pace. In the light of this situation, it is essential to periodically launch the up-to-date Physics Curriculum to stay at par with the international standards.

Curriculum development is a highly organized and systematic process and involves a number of procedures. Many of these procedures include incorporating the results from international research studies and reforms made in other countries. These studies and reforms are then related to the particular subject and the position in Ethiopia so that the proposed curriculum may have its roots in the socio-economic setup in which it is to be introduced. Information obtained at this level can be used as input for launching, revising and/or closing down of existing programs in the university. Hence, unlike a machine, it is not possible to accept any curriculum in its entirety. It has to be studied thoroughly and all aspects are to be critically examined before any component is recommended for adoption. The MSc programme in Physics were designed to offers two-year program in regula and two and half(including two summer program) year program in extension program. The team of Physics department initiated the opening of MSc program in MWU by consulting with post graduate , Natural and Computational College ,academic vice precedent office three year ago in 2014. However,the University Senate blessed the opening of MSc Physics in summer program three year ago, holding up the opening of regular program that requiring further assessment required. Based upon this assumption, the team of physics department in the month of November(2017) collaborated with School of Graduate and Assurance and Audit directorate made the need assessment for possible launching of MSc in physics both in regular and extension program. More over the team assessed other universities (Hawassa, Jima, and Welega) experience on the conditions of past and present scenario of their programs. The team thanked the department heads of physics department for unwavering collaborations they provide us. The experience we shared and the need assessment would make outline for possible launching of MSc in physics both in regular and extension program in two specialization courses. These are:

1. Master of Science in Physics(Quantum Optics)

2. Master of Science in Physics(Solid State Physics)

ABSTRACT

Abstract

The role of Physics in the economic development of a nation and the prosperity of its people is very important and it is imperative for a nation to train its workforce in contemporary Physics. At the same time, knowledge is expanding rapidly and new courses and disciplines are emerging at a fast pace. In the light of this situation, it is essential to periodically launch the up-to-date Physics curriculum to stay at par with the international standards. Base on the need assessment, at present manpower status and future objective of the university, a physics department is planning to launch Master of Sciences in Physics(Quantum Optics and Solid State Physics in two year regular and three years in extension/weekend programs in the coming academic year(2011 E.C/2018/19). Master of Science in Physics in summer program was already opened three years ago(2014) with Harmonized Curriculum prepared by Ministry of Education

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Chapter 1

Introduction

Across the world, science is increasing being viewed as a subject of life-long utility to all human kind. Amore science literature populace is perceived as being better equipped to contribute to sustainable economy development and to the social welfare. Therefore, factor which contribute to high achievement in science are of concern to decision-maker in the country. Among science field , applied physics is the most indispensable part playing wide role in supporting and speeding up to understand or conceptualize other academic fields like engineering technology for achieving the set goal of the country for drastic economic transformation[1,2]. However the quality of physics among science education with skill full qualified both in secondary and tertiary levels have been a huge challenge for Ethiopia Education system for many years[2]. Developing countries like Ethiopia are under huge pressure from international community to implement on of conventions called United Nation Millennium Development Goals, with high priority to achieving quality science education by 2015[2,3]. Investment in human capital through training and educational programs is seen as determinant of the increase in the value of the capital.[4]To achieve the desired goal, the quality and pedagogy of science educations including physics with its application with trained human power have to be improved through not only upgrading the existing field of studies but also by launching new MSc program based on the need assessment of the community.

Physics, as one of the fundamental sciences, is concerned with the observation, understanding and prediction of natural phenomena and the behavior of man-made systems. It deals with profound questions about the nature of the universe and with some of the most important practical, environmental and technological issues of our time. The scope of Physics is broad and encompasses mathematical and theoretical investigation, experimental observation, computing technique, technological application, material manipulation and information communication and processing

Physics seeks simple explanations of physical phenomena based on universal principles stated in concise and powerful language of mathematics. The principles form a coherent unity, applicable to objects as diverse as DNA molecules, neutron stars, super-fluids, and liquid crystals and the application of fiber optics and photopic technology in communication [4]. Findings in Physics have implications in all walks of life ranging from the way we perceive reality to gadgets of everyday use.

1.1 Goal

The goal of Postgraduate program in Physics is to provide a theoretical and practical knowledge base for those with an interest in developing an inter-disciplinary approach to problem solving in different areas of physics, and in particular those seeking employment as a lecturer in university/college environments, upgrading preparatory schools, aviation, mining, medical physicists or as biomedical engineers in hospital, industry and so on. The program has been appropriately designed to equip the students with skills suitable for application in the wider job market as well as for sustainable exploitation of the vast natural resource potential for national development. It also used to equip the students to become effective teachers and researchers in Physics, to contribute to the needs of the society by providing an environment of learning and knowledge creation through academic rigor and innovations. The study covers both theoretical and experimental Physics, through lectures, tutorials, and laboratory experiments.

1.1.1 Rationale of the program

It is now widely believed that science and technology are very essential for socio-economic developments. In line with this, presently there is a great demand for Physics graduates. We strongly believe that this MSc program will meet the demand partially.

1.1.2 Program Objectives and Outcomes:

The Master Degree program in Physics in the Physics Department at Madda Walabu University is designed to provide the students with advanced knowledge in core subjects in physics and prepare them to pursue their higher studies towards the Ph.D. degree in physics, if they so chose. In addition, this program will equip the students with the training and skills in experimental and numerical methods to solve advanced problems in Applied Physics. Graduates from this program are expected to be able to meet the demands of the ministry of education in having well trained and skilled professionals to promote their education program, and have the training and background to join and help existing and new industries and quality teaching profession in the country. The Postgraduate Course in Physics is the final formal training for students and rigorous training which is required to provide in depth knowledge of the subject. Hence the regular class room teaching is supplemented with tutorials, brain storming ideas and problem solving efforts, pertaining to each theory and practical course. Foundation courses and seminars are introduced to help the students to achieve holistic development and to prepare them to face the world outside in a dignified manner.

1.1.3 Specific Objectives:

- Establish a firm foundation for scientific and technological development by providing basic research techniques in physics. communication, research, and teaching at University/College levels and Secondary Schools
- Pursue higher level training in different fields of physics
- Develop skills of numerical manipulation and statistical analysis of data using sophisticated computer software.
- To provide, through well designed studies of theoretical and experimental Physics, a worthwhile educational experience for all students.

- To acquire basic knowledge in the study fields thrust areas like, Quantum Mechanics, Computational & Mathematical Physics, or solid state Physics, etc
- To develop manpower in teaching physics at secondary School, College and the tertiary level and to conduct research in physics.
- To produce high level research manpower in physics.

1.2 Eligibility for Admission

- The candidates who have passed B.Sc. or B.Ed degree with major in physics from recognized Universities.
- An applicant seeking admission to M.Sc. physics must appear in an entrance examination conducted by Department of Physics and the applicant should pass the exam decided by DGC.
- Candidates must meet the criteria set by the School of Graduate studies (SGS) in addition to the written entrance examination administered by the department.
- Candidates must be capable of presenting a letter of sponsorship from their organization or should be able to present an equivalent amount of financial source for the whole study period of the program.
- Candidates should be able to provide two letters of recommendation.
- Candidates preferably be with a working experience (Optional)
- The applicant who couldn't appear for the entrance exam and/or fail to obtain a minimum qualifying score will not be given admission.

1.3 Professional Profile

The path through the physics universe will include theoretical instruction experiencing in project and thesis work in the field. Graduate students in physics is specialized in one of three sub-fields such as quantum optics,solid-state physics,and computational physics. As a graduate student in physics he will achieve the academic qualifications to independently participate in modern research and the communication of results. There are plenty of themes and courses to choose from, so you will determine your own course of study to a large degree.

1.3.1 Competence Description

The candidate will achieve the following basic profiles as staying in the programmes

- attaining the basic physical principles governing a physical system.
- working with mathematical and numerical models to describe physical reality.
- attaining learning to work with experimental setups and become familiar with the latest technologies.
- work with the processing,interpretation of experimental and numerical data.
- read current research literature and be capable of contributing to the research.

1.3.2 Career Opportunities

Unemployment among physics graduates is quite rare indeed and most physicists find work in a variety of settings.

- it is possible for all physicists to qualify for teaching at high school level, college,and territory level .

- Working on research institutions, private and public organization
- From need assessment result , most master graduated in physics will be employed in educational sectors like Preapratory schools, College and Universities)
- Ethiopian Radiation Protection Authority
- Quality & Standards Authority Of Ethiopia
- Ethiopia Space and Technology Institute(ESTI)

1.3.3 VISION, MISSION AND VALUES OF THE MSc Program in Physics

1. **Vision**: To be a World Class Centre for Research, Education and Innovation in the Sciences.
2. **Mission**: To provide globally competitive and high quality academic programmes in Science, to produce employable graduates through excellence in teaching and research, to undertake national and international research in priority fields including through collaborative partnerships and industry linkages, and to promote the application of science for the benefit of society
3. Mission Statements:
   - Teaching and Knowledge Transfer:teaching postgraduate programmes relevant to national, regional and international needs;
   - inward knowledge transfer through inter alia, industrial advisory boards and international scholar exchange programmes.
4. Research and Knowledge Creation
   - research in the major science disciplines

- National and international Centres of Competence in research clusters which are multi-disciplinary and focused on strategic research priorities;
- collaborative partnerships and industry linkages
- applied research for external clients including industry and government.

5. Application of Science for the Benefit of Society:

- providing expert advice and work through consultancy, membership of national and regional committees, engagement with industry,government, communities, and academic bodies;
- technology transfer from research via incubator facilities;
- partnerships in specific projects for strategic goals with external agencies both national, regional and international
- community engagement through academic endeavours which are responsive to local community needs.
- Application of physics in industry,theology and social transformation

6. Values

- Benchmark itself to international standards of scientific practice and critical thinking such that we produce world-class science;
- Actively encourage and respect the right of all scholars, staff and students to engage in critical inquiry, independent research, intellectual discourse and public debate in a spirit of responsibility and accountability, in accordance with the principles of academic freedom;
- Promote access to learning that will expand educational and employment opportunities for all;

- Embrace its responsibility to support and contribute to national and regional development, through the generation and dissemination of knowledge, the production of socially-responsible graduates and the application of science for the benefit of society;
- Conduct itself according to the highest ethical standards, and provide education that promotes an awareness of sound ethical practice and academic integrity;
- Conduct itself according to the highest ethical standards, and provide education that promotes an awareness of sound ethical practice and academic integrity;
- Ensure effective governance through broad and inclusive participation, representation, accountability, and transparency that serves as an example of true collegiality;
- Acknowledge the value of the individual by promoting the intellectual, social and personal well-being of staff and students through tolerance and respect and by fostering the realisation of each person's full potential;
- Aspire to the highest quality in all its endeavours

1.4 Resource Profile and facilities

1.4.1 Human Resources

The Department of Physics has several members that can potentially contribute for effective development of MSc Physics program in MWU to attain the motto, vision,mission and value of the department within the university.

1.4.2 Laboratory,Computer, References

The two courses can be treated by two type of Laboratories. They are

1. Computational Laboratory: This laboratory needs desk top computers,computer softwares that are essential for modelling and simulating physical phenomena to be visualized in a

Table 1.1: Physics Staff Members on Active Duties

No.	Name	Sex	Rank	Specialization	Remark
1	Dr. Cherinet Seboka	M	Ass. Professor	Radio Physics/ Metamterial	
2	Dr. Getachew Asmelash	M	Ass. Professor	Quantum Optics	
3	Mr. Said Endris	M	Lecturer	Computational Physics	
4	Mr. Adem Beriso	M	Lecturer	Solid State Physics	
5	Mr. Getinet Tessema	M	Lecturer	Nuclear Physics	
6	Mr. Jeylan Hussen	M	lecturer	Environmental Physics	
7	Mr. Umer Sherfedin	M	lecturer	Physical Electronics	
8	Mr.IEbrahim Kedir	M	lecturer	Physics Education	
9	Mr. Shiferaw Wami	M	Lecturer	Material Physics	
10	Mr. Abebe Adugna	M	Lecturer	Quantum Optics	
11	Ms. Fantu Worqu	F	Grad. Assistance	B.Sc(Physics)	
12	Mr.Andinet Teshome	M	Lab. Technician	B.Sc(Physics)	
13	Ms.Sisay Dame	F	Lab.Technician	Diploma(10+3)	

Table 1.2: Physics Staff Members on Study Leaves

No.	Name	Sex	Study For	Specialization	Study Place	Expected Date of Coming
1	Abdulahi Abdela	M	PhD	Solid State Physics	AAU	August 2019
2	Zelalem Abebe	M	PhD	Materials Science & Engineering	Beijing	August 2019
3	Abdisa Kawo	M	PhD	Space Science	ESST(AA)	August 2020
4	Endalemaw Dessie	M	PhD	Physics Education	AAU	Agust 2021

smooth approach. These soft wares are Matlab,CST Microwave, Mathematica, M C++, etc. These softwares are potentially essential in modelling both analytically and numerically to describe the physical phenomena in computation. For beginning the program, a minimum 15 desk top computer fully installed with wired internet.

2. Equipment Laboratory:There are some laboratory equipments that can initiate future program to be opened requiring intensive laboratory works . For instance; measuring light speed, spectrometer, Diffraction. More work has to be done soon to fulfill the remaining the apparatus in collaborating with College and University Official.

1.5 Grade & Grade Point Average (GPA):

The performance of a student shall be made on a 12 point scheme ranging from 0 to 4 grades. Students shall receive their semester grades and academic transcript grades only in letter grades and GPA scores. The percentage equivalent of the grade and GPA is as follows in table1.3:

Table 1.3: Letter Grading System with the corresponding Grade Points

Raw Mark	Grade point	Grade Letter
[90-100]	A+	4.00
[85-90)	A	4.00
[80-85)	A-	3.75
[75-85)	B+	3.5
[70-75)	B	3.00
[65-70)	B-	2.75
[60-65)	C+	2.5
[50-60)	C	2.00
[<50)	F	0

1.6 Quality Assurance

The quality of the program shall be audited in accordance with the guidelines set by Higher Education Relevance and Quality Agency. However, the Physics Department will monitor and maintain the quality of the program according to the Quality Assurance Standard set by the University. To this effect, the Physics Department will:

- Ensure that the contents of the courses are covered
- Ensure that exams, tests, assignments, and projects etc. are properly set and conducted
- Ensure that appropriate technology is employed in the teaching-learning process
- Ensure that appropriate and recent text books are used for the courses and provide enough reference books for each course
- Evaluate the courses at the end of each semester based on the feedback obtained from the instructors, tutors, and students, so as to make the course more relevant
- Conduct short term courses and seminars for staffs members in order to make use of modern methodology and IT
- Make sure that tutorial classes are well organized, relevant exercises, home works, and assignments are carefully set to enhance and strengthen the students' ability to solve practical problems and understand the underlying theory

On the other hand, At the end of each semester miner survey will be conducted so that students will be asked to fill a questionnaire about the quality of the lectures. The results of the exams will also be analyzed at the end of each semester. Students will also be asked to suggest their opinion to improve the course delivery and the assessment methods. In order to collaborate with students to improve the teaching-learning mode, the representatives of students will be allowed to attend the department meeting whenever needed.

1.7 Program Requirements

1.7.1 Admission Requirements

- Possess BSc./ BEd. degree in Physics from any recognized University / College or BSc degree in Equivalent /other discipline (like a degree from an engineering school, Mathemtics)
- The applicant with BEd degree in physics or equivalent may be given admission provided that he /she will take remedial courses proposed by the Department Graduate Committee (DGC).
- The applicant must score a result for the MSc entrance examination in the range that will be fixed by the DGC.
- The applicant should satisfy the general admission requirements set by the University Senate Legislation.

1.7.2 Course Requirements

The courses in the program are classified in two parts. The first part covering 19 Cr.Hrs/44 ECTS in the first year of regular program are termed as as general or compulsory courses. Whereas the second part are classified under field of specialization covered in 2nd year. The total credit of the area of specialization, seminar courses and selective course are supposed to a 10 credit hour/28ECTS. The graduate thesis is registered to 6 credit hours/ 28 ECTS. However, the student must be registered to the thesis in the first semester of the second academic year with zero credit hours with the same work load indicated by star symbol"*" . The same assumption is also applicable to extension or weekend program. Accordingly, the total credit hours needed to run the program with including Graduate Thesis work in Extension(Weekend) program is 35 credit hours/100 ECTS. Regular program covers two academic years whereas Extension(Weekend) Program covers Three solid academic years. When the requirement is summarized as minimum Course load the graduate accomplishing in Master of Science in Physics with including graduate Thesis is 35 credit hours or 100 ECTS

1. 19 Cr.Hrs/44 ECTS of Compulsory Courses
2. 10 Cr.hrs/28 ECTS of Specialization, Seminar and Special topics Courses
3. 6 Cr.hrs/28 ECTS of Graduate Thesis

1.7.3 Graduation Requirements

The masters of Science Degree in physics is awarded to candidates who will fulfill the following requirements

- the graduates must learn all the required courses and pass all examinations in courses offered in the program and score a minimum CGPA of 3.00,
- the graduate student must complete a 35 credit hours 100ECTS courses/ including a 6 credit hours /28 ECTS Graduate Thesis o
- the graduate student must undertake a graduate thesis on approved topic under the supervision of advisor(s),
- the graduate student must successfully defend the thesis score a minimum mark of ”satisfactory” grade
- the thesis or the project work shall constitute individual effort to carry out a fairly adequate investigation of a selected problem,
- if a student has a grade of C+ and/or C in more than one course, he/she is subjected to repeat the course,
- the examination committee shall be composed of two examiners (external and internal examiners) and the chairman for thesis defense,

- the graduate student must also fulfil all the graduation requirements of the University,
- no "F" in any course.

1.8 Duration of the study

1.8.1 Regular Program

The Regular MSc program will be offered in four semesters or two year program. A candidate can be allowed to continue to a maximum of three years under special cases in regular program. The extension should be recommended by the Department Graduate Committee (DGC) and approved by the Academic Commission. The program shall take two years subject to the possible extension for one years in line with the University's Senate Legislation.

1.8.2 Extension/ Weekend

The duration of the extension program is six semesters. A candidate can be allowed to continue to a maximum of eight semesters under special cases. The extension should be recommended by the Department Graduate Committee (DGC) and approved by the Academic Commission.

Table 1.4: Compulsory(Core) Module

Module No.	Course No.	Course Title	Cr. Hrs/ECTS
Phys- M600	Phys 601	Mathematical Methods of Physics	3/6
	Phys 602	Research Method in Physics	1/7
	Phys 603	Thermodynamics and Statistical Mechanics	3/6
	Phys 604	Classical Electrodynamics	3/6
	Phys 605	Classical Mechanics	3/6
	Phys 606	Quantum Mechanics	3/7
	Phys 608	Computational Physics	3/6
		Total	19/44

Table 1.5: List of Specialization Course

	Module No.	Course No.	Course Title	Cr. Hrs/ECTS	Remark
1	Phys-M711	Phys 711	Quantum Optics I	3/7	Double Cr.hrs
		Phys 713	Quantum Optics II	3/7	in 2 months
2	Phys-M721	Phys 721	Solid State Physics I	3/7	≫
		Phys 723	Solid State Physics II	3/7	
3	Phys-731	Phys 731	Seminar	1/7	
		Phys 732	Selected Topics	3/7	
4	Phys-M742	Phys 742	Graduate Thesis	6/28	

1.9 Module, courses list and Sequencing, and description List of Modules and Courses

1.9.1 Compulsory(core) Module and Courses

1.9.2 Specialization Modules and Courses

1.10 Course coding and sequence

All courses are coded "Phys" followed by three digits (i.e., Phys xxx):

✠ The first digit indicates the level of the course (6 for first year and 7 for second year of the graduate course).

✠ The middle(second) digits indicate the course type i.e.. various streams of Physics courses, i.e.,

- 0 - Compulsory (Common) Courses
- 1 - Quantum Optics
- 2 - Condensed Matter (Solid State)Physics
- 3- Special Topics and seminar
- 4-Graduate Thesis

✠ The last digits stand for semester in which the course is to be delivered (with the exception of Graduate Thesis), i.e.,

- ODD last digit courses are to be delivered during the first semester or Odd semesters (Regular Program).
- EVEN last digit courses are to be delivered during the second semester or Even semesters (Regular Program).

1.10.1 Module Coding

Modules are coded "Phys" appended by "-M" followed by three digits

✠ The first digit indicates the level of the module:

✓ 6 for first year modules and

✓ 7 for second year modules.

✠ The middle digit indicates the serial number of the module in the program 1, 2, 3, . . .

✠ The last digit indicates the type of the module:

✓ 0 for compulsory modules,

✓ 1 for specialization modules, and

✓ 2 for Graduate thesis.

1.11 Modes of Delivery(Modes of Teaching and Learning)

The MSc study program in Physics provides the student with more advanced courses, and begins the process of specialization. The students will through advanced topics, with lectures, exercises and laboratory work, as well as an independent study (the Master's degree thesis) acquire knowledge of relevant working methods in research, industry, administration and teaching.

Lectures give a formal introduction to the scientific methods in physics and advance thinking in a scientific way. They help in acquiring the basic knowledge and understanding the fundamental physics concepts.

Exercise classes complement the lectures and provide essential help for the understanding and practical application of a lecture's content. Through them, the student will practice and apply the acquired principles and mathematical techniques

Computational Laboratories enabling students to develop skills on computational approach for solving physics problems to be adopted, which is distinct from, and complimentary to, traditional experimental and theoretical approaches.

- ☐ focusing on the science aspects of computational Laboratories for Numerical analysis
- ☐ providing students hands-on experience in designing and running computer for computer simulation simulations
- ☐ playing a role similar to that of laboratory courses for experiment
- ☐ developing Project-based computational physics for solving real problems

Preparation of the Master thesis , under the supervision of an experienced researcher, is the actual starting point of scientific research.

- ☐ The mode of teaching-learning process will involve lectures and student participation.
- ☐ The lectures provide the means to give concise, focused presentation of the subject matter of the course. When appropriate, the lectures will be supported by the distribution of written material, or by relevant links on websites.

- ☐ Student participation involves presentation on specific topics.
- ☐ Regular problems and exercises will give students the chance to develop their theoretical understanding and problem solving skills.
- ☐ Students will be able to obtain further help in their studies by approaching their lecturers, either after lectures or at other mutually convenient times.
- ☐ Computational Physics will involve lectures and laboratory sessions.

1.12 Modes of Evaluation

Student performance will be assessed through examinations, presentations, problems, and exercises. The examinations, problems and exercises will provide the means for students to demonstrate the acquisition of subject knowledge and the development of their problem-solving skills. The problems and exercises provide opportunities for students to gauge their progress and for the instructor to monitor the progress of his students throughout the duration of the course(s) in the modules. For Computational Physics and Experimental Physics courses student evaluation will also involve practical laboratory examinations.

1.12.1 Assessment and Evaluation

The following assessment processes shall be followed:

I Course Work

1. Quizzes
2. Assignment
3. Written Exams
4. Presentations

II Laboratory and Computational Courses

1. Laboratory and Field Reports
2. Practical Examinations
3. Written Examinations

III Seminars and Thesis

1. Quality of paper presented
2. Originality of the paper(Only for Thesis)
3. Way of presentation
4. Defending the work presented

1.13 Dissertation

- A student who finalizes his thesis work , should give a pre-presentation (Mock defence) in front of departments staffs before the final VIVA examination. The advisor should give endorsing signal before presenting his work in the two presentations. The mock defence is believed to give empowering comments before he/she presents his/her research before the VIVA session. The final VIVA examination day will be fixed by the department head in collaboration with school of post graduate. The examination commute during viva presentation is composed of internal examiners,external examiners and the chair person which is a scheduled exam with internal and external examiners. The head of the evaluation committee will be , M.Sc. program coordinator and/or head of the department. The write up of thesis should be in a recommended format.

1.14 Specialization Area And Courses

The MSc. program is delivered in two specialization areas listed below:

1. Quantum Optics

2. Solid State Physics

A graduate student is expected to choose one of the following areas of specialization.

Table 1.6: Specializations

	Specialization Area	Module No.	Course Code	Course Title	Cr. Hrs/ECTS
1	Quantum Optics	Phys-M711	Phys 711	Quantum Optics I	3/7
			Phys 713	Quantum Optics II	3/7
		Phys-M731	Phys 731	Seminar	1/7
			Phys 732	Selected Topics	3/7
		Phys-M742	Phys 742	Graduate Thesis	6/28
2	Solid State Physics	Phys-M721	Phys 721	Solid State Physics I	3/7
			Phys 723	Solid State Physics II	3/7
		Phys-M731	Phys 731	Seminar	1/7
			Phys 732	Selected Topics	3/7
		Phys-M742	Phys 742	Graduate Thesis	6/28

1.15 Degree Nomenclature

Name of the Master Program is written as ”Master of Science in physics (*Course of Specialization*” in both English and Amharic languages.

1. **The Degree of Master of Science in Physics (Quantum Optics)**
   የሣይንስ ማስተር ዲግሪ በፊዚክስ (በኳንተም ኦፕቲክስ)
2. **The Degree of Master of Science in Physics (Solid State Physics)**
   የሣይንስ ማስተር ዲግሪ በፊዚክስ (በሶሊድ ስቴት ፊዚክስ)

Chapter 2

Course Break Down for Regular Program and Extension/Weekend Program

2.1 Regular Program

From the need assessment and present manpower status of the department and future objective of the university, a two year regular program is to be launched in Master of Science in Physics(Quantum Optics, Solid State physics) where theoretical physics are prevailed. In the first year of the regular program, all courses are the same. In the second of the program,students are streamed to specialization or concentration area.

2.1.1 Specialization in Quantum Optics(Regular)

The graduate thesis is registered to 6 credit hours/ 28 ECTS. However, the student must be registered to the thesis in the first semester of the second academic year with zero credit hours indicated by star symbol"*" with the same work load as second semester. The same assumption is also applicable to extension or weekend program.

Table 2.1: Course Breakdown for in Quantum Optics(Regular)

N0.	Course Code	Course Title	Cr.Hrs/ECTS	Sem.	Sub Tot. Cr.Hrs/ECTS
1	Phys 601	Mathematical Method of Physics	3/6	I	
2	Phys 603	Thermodynamics and Statistical Mechanics	3/6	I	9/18
3	Phys 605	Classical Mechanics	3/6	I	
1	Phys 606	Quantum Mechanics	3/7	II	
2	Phys 608	Classical Electrodynamics	3/6	II	
3	Phys 602	Research method in Physics	1/7	II	10/26
4	Phys 608	Computational Physics	3/6	II	
1	Phys 731	Seminar	1/7	III	
2	Phys711	Quantum Optics I	3/7	III	
3	Phys 713	Quantum Optics II	3/7	III	7/21
4	Phys 742	Graduate Thesis	*	III	
1	Phys 732	Selective Topic	3/7	IV	9/35
2	Phys 742	Graduate Thesis	6/28	IV	
		Total Cr.Hrs	35/100		35/100

Table 2.2: Course Breakdown of Specialization in Solid State Physics(Regular)

N0.	Course Code	Course Title	Cr.Hrs/ECTS	Sem.	Sub Total Cr.Hrs/ECTS
1	Phys 601	Mathematical Method of Physics	3/6	I	
2	Phys 603	Thermodynamics and Statistical Mechanics	3/6	I	9/18
3	Phys 605	Classical Mechanics	3/6	I	
1	Phys 606	Quantum Mechanics	3/7	II	
1	Phys 608	Classical Electrodynamics	3/6	II	
2	Phys 602	Research method in Physics	1/7	II	10/26
3	Phys 608	Computational Physics	3/6	II	
1	Phys 731	Seminar	1/7	III	
2	Phys 721	Solid State Physics I	3/7	III	
3	Phys 723	Solid State Physics II	3/7	III	7/21
4	Phys 742	Graduate Thesis	*	III	
1	Phys 732	Selective Topic	3/7	IV	
2	Phys 742	Graduate Thesis	6/28	IV	9/35
		Total Cr.Hrs	35/100		35/100

2.1.2 Specialization inSolid State Physics(Regular)

2.2 Extension(Weekend Program)

The extention/weekend program lasts for three academic years.

2.2.1 Specialization in Quantum Optics(Extension/Weekend)

Table 2.3: Course Breakdown of Specialization in Quantum Optics(Weekend)

N0.	Course Code	Course Title	Cr.Hrs/ECTS	Sem.	Sub Total Cr.Hrs/ECTS
1	Phys 601	Mathematical Method of Physics	3/6	I	
2	Phys 603	Thermodynamics and Statistical Mechanics	3/6	I	6/12
1	Phys 605	Classical Mechanics	3/6	II	
2	Phys 606	Quantum Mechanics	3/7	II	6/13
1	Phys 604	Classical Electrodynamics	3/6	III	
2	Phys 602	Research method in Physics	1/7	III	7/19
3	Phys 608	Computational Physics	3/6	III	
1	Phys 731	Seminar	1/7	IV	
2	Phys711	Quantum Optics I	3/7	IV	
3	Phys 713	Quantum Optics II	3/7	IV	7/21
1	Phys 732	Selective Topic	3/7	V	3/7
2	Phys 742	Graduate Thesis	*	V	
1	Phys 742	Graduate Thesis	6/28	VII	6/28
		Total Cr.Hrs	35/100		35/100

2.2.2 Specialization in Solid State Physics(Extension/Weekend)

Table 2.4: Course Breakdown of Specialization in Solid State Physics(Weekend)

N0.	Course Code	Course Title	Cr.Hrs/ECTS	Sem.	Sub Total Cr.Hrs/ECTS
1	Phys 601	Mathematical Method of Physics	3/6	I	
2	Phys 603	Thermodynamics and Statistical Mechanics	3/6	I	6/12
1	Phys 605	Classical Mechanics	3/6	II	
2	Phys 606	Quantum Mechanics	3/7	II	6/13
1	Phys 604	Classical Electrodynamics	3/6	III	
2	Phys 602	Research method in Physics	1/7	III	7/19
3	Phys 608	Computational Physics	3/6	III	
1	Phys 731	Seminar	1/7	IV	
2	Phys721	Solid State Physic I	3/7	IV	
3	Phys 723	Solid State Physics II	3/7	IV	7/21
1	Phys 732	Selective Topic	3/7	V	3/7
2	Phys 42	Graduate Thesis	*	V	
1	Phys 742	Graduate Thesis	6/28	VII	6/28
		Total Cr.Hrs	35/100		35/100

Chapter 3

Course Description

3.1 Phys 601:Mathematical Methods of Physics (3Cr.Hrs/ 6 ECTS)

Instructional Objective

1. To develop knowledge in mathematical physics and its applications.

2. To develop expertise in mathematical techniques those are required in physics.

3. To enhance problem solving skills.

4. To give the ability to formulate, interpret and draw inferences from mathematical solutions.

3.1.1 UNIT I: VECTOR ANALYSIS

Vectors and Vector Spaces - Definition - Transformation of Vectors - Rotation of the Coordinate Axes - Invariance of the Scalar Product Under Rotations - Gradient - Divergence - and Curl of Vectors - Physical Interpretation - Vector Integration - Line- Surface and Volume Integrals - Gauss' theorem - Stokes' theorem - Dirac Delta Function - Integral Representations- Vector Analysis in Curved Coordinates - Expression for Gradient- Divergence and Curl in Spherical Polar Coordinates- Tensors - Contravariant and Covariant tensors- Definition of tensor of rank two.

3.1.2 UNIT II: DIFFERENTIAL EQUATIONS

as a special solution - Second Order Differential Equations with constant coefficient - Bessel polynomials - Legendre Polynomials - Hermite and Laguerre polynomials and their differential equations- Generating functions- Recurrence relations- Orthogonality of functions - One dimensional Green's functions - Sturm Liouville's type equation.

3.1.3 UNIT III: COMPLEX VARIABLES

of a complex variable - Single and multivalued functions- Analytic functions - Cauchy Riemann Conditions - Cauchy's Integral Theorem - Cauchy Integral Formula- Taylor and Laurent Expansions - Laurent Series - Essential Singularities - Removal Singularities - Isolated Singularities - Poles and Branch Points - Conformal Mapping - Translation - Rotation and Inversion.

3.1.4 UNIT IV: MATRICES & GROUP THEORY

Matrices - Basic Definitions - Inverse of a matrix - Direct Product of matrices - Orthogonal - Hermitian - Unitary and Normal Matrices- Eigenvalues and Eigenvectors - Degenerate eigenvalues - Introduction to Group Theory- Definition of a Group- Homomorphism- Isomorphism - Rotations as an example- Generators of Continuous Groups- Rotation Groups SO(2) and SO(3)- Rotation of Functions- Discrete Groups.

3.1.5 UNIT V: INTEGRAL TRANSFORMS

Introduction of Integral Transforms- Fourier Transforms - Definition- Linearity- Development of the Fourier Integral- Fourier Cosine and Sine Transforms- Application to heat flow and wave equations- Convolution theorems- Parseval's Relation- Momentum Representation- Laplace Transforms -Definition- Linearity- Shifting formula and Inverse Laplace transforms.

TEXTBOOKS

- Arfken and Weber, Mathematical Methods for Physicists, 6th Edition, Elsevier, 2012.

- Murray R. Spiegel, Schaum's Outline of Advanced Mathematics for Engineers and Scientists, 1st Edition, McGraw Hill, 2009.

References

- Gupta B. D., Mathematical Physics, 4th Edition, Vikas Publishing House, 2009.
- Sadri Hassani, Mathematical Physics: A Modern Introduction to Its Foundations, 2nd Edition, Springer, 2013.
- Foundations of Mathematical Physics, Sadri Hassani, Allyn and Bacon Press.
- Mathematical Methods, G. Erfken, Academic Press Inc., London.
- Mathematical Methods for the Physical Sciences, Roel Snider.
- Chattopadhyay P. K., Mathematical Physics, 1st Edition, New Age International, 2009.

3.2 Phys 605:Classical Electrodynamics(3Cr.Hrs/ 6 ECTS)

INSTRUCTIONAL OBJECTIVES

- To make the student understand the principles of electrostatics and magnetostatics.
- To enable the student to explore the field of electrodynamics.
- To make the student understand the basic concepts in Electromagnetic wave and radiation
- To allow the student to have a deep knowledge of the fundamentals of Electromagnetism.

3.2.1 UNIT I : ELECTROSTATICS

Coulomb's Law - Electric Field- Gauss's law of electrostatics - Sphere- Shell- Plane- line- Electrostatic Potential- Multipole expansion- Work and energy in electrostatics-Poisson's Equation - Laplace equation-

Boundary condition and uniqueness theorem- Method of Images - Classic image problem- Induced Surface charge- Separation of variables - Electrostatics in matter - Polarization - Free charges - Bound charges - Linear isotropic homogenous medium.

3.2.2 UNIT II: MAGNETOSTATICS

Lorentz force law-Biot-Savart law-Divergence and curl of B-Magnetic field of steady currents - Ampere's law -Magnetic field of a straight wire - Magnetic field of a solenoid-Comparison of electrostatics and magneto statics- Magnetic potential -Magnetization- Fields of a magnetized object- Bound currents- Physical interpretation of bound currents- Magnetic field inside a matter.

3.2.3 UNIT III : ELECTRODYNAMICS AND ELECTROMAGNETIC WAVES

Electromotive force- Ohm's law- Electromagnetic induction - Faraday's law - Lenz law- Maxwell's equations -Displacement current- Electromagnetic waves- In vacuum- In dielectric media- Skin depth-Poynting vector- Boundary conditions- reflection and transmission at normal incidence- Reflection and transmission at oblique incidence - Fresnel's equations - Perpendicular polarization - Parallel polarization.

3.2.4 UNIT IV: POTENTIALS AND RADIATION

Continuous distribution - Retarded scalar and vector potentials-Jefimenko's potential - Point charges - Lienard-Wiechert potentials for a moving point charge-Electric and magnetic fields of a moving point charge-Electric dipole radiation-Magnetic dipole radiation - Power radiated by a point charge - Radiation reaction - The physical basic of a radiation reaction.

3.2.5 UNIT V: RELATIVISTIC ELECTRODYNAMICS

Potential formulation of electrodynamics - Gauge transformations-Coulomb gauge and Lorentz gauge-Introduction to special theory of relativity - Magnetism as a relativistic phenomenon- Field transformation-

Field Tensor- Electrodynamics in Tensor notation - Relativistic potentials - The four potential of a moving charge - The invariance of the equations of electrodynamics.

TEXT BOOKS

- Griffith D.J, Introduction to Electrodynamics, 4th Edition, Prentice Hall of India, 2012.
- John David Jackson, Classical Electrodynamics, 3rd Edition, John Wiley & Sons Ltd, 1998.
- Murugeshan.R, Electricity and Magnetism, 7th Revised Edition, S. Chand, 2008.
- Laud B.B, Electromagnetics. 2nd Edition, New Age International Publication, 2005.

References

- Kip A.F, Fundamentals of Electricity & Magnetism, International Student Edition, Mc Graw, Hill, 2008.
- Feynman, Leighton and Sands, The Feynman Lectures on Physics, Volume 2, Narosa Publishing house,2008.
- Berkeley, Physics Course: Electricity & Magnetism, 2nd Edition, Tata Mc Graw Hill, 2007.
- Stratton and Julius Adams, Electromagnetic Theory, Vol. 33, John Wiley & Sons, 2007.

3.3 Phys 603:Thermodynamics and Statistical Mechanics (3Cr.Hrs/ 6 ECTS)

Instructional Objective

1. The course is to understand the basics of thermodynamics and Statistical systems
2. Understand the various laws of thermodynamics.
3. Acquire the knowledge of various statistical distributions.

4. To comprehend the concepts of enthalpy, phase transitions and thermodynamic functions.

3.3.1 UNIT I: BASIC ELEMENTS OF STATISTICAL MECHANICS

Foundations of statistical mechanics-Bridging microscopic and macroscopic behavior -Contact between statistics and thermodynamics- Free energy-Partition function of an ideal monatomic gas -Partition function of a diatomic molecule- Classical ideal gas- Entropy of mixing and Gibb's paradox-Sackur -Tetrode equation-The semi-classical perfect gas- Specific heat of solids -Saha's ionization formula.

3.3.2 UNIT II: THEORY OF ENSEMBLES

Ensembles- Microcannonical ensemble-Phase space- Trajectories and density of states-Liouville's theorem-Canonical ensemble -Thermodynamic properties of the canonical ensemble-Evaluation of the total partition function-Partition function in the presence of interactions-Fluctuation of the assembly energy in a canonical ensemble- Grand canonical ensemble- The grand partition function and its evaluation- Fluctuations in the number of systems-The chemical potentials in the equilibrium state.

3.3.3 UNIT III: STATISTICAL DISTRIBUTIONS AND APPLICATIONS

Maxwell-Boltzmann distribution- Determination of undetermined multipliers β α-equipartition of energy-Bose-Einstein statistics-Bose-Einstein condensation-Fermi-Dirac statistics- The electron gas-White dwarfs and their limiting mass, statistical (Thomas - Fermi) model of atom-Comparison of the 3 statistics.

3.3.4 UNIT IV: LAWS OF THERMODYNAMICS AND THERMODYNAMIC PROCESSES

Ideal gases: Equation of state- Internal energy- Specific heats -Entropy- isothermal and adiabatic processes-Compressibility and expansion coefficient-.Real gases: Deviation from the ideal gas equation-Zeroth and first law of thermodynamics-Reversible and irreversible processes- Carnot theorem- Second law of thermodynamics-Clausius inequality- Entropy changes in reversible and irreversible processes- Temperature-entropy diagrams-Unavailable energy-Thermal death of the universe.

3.3.5 UNIT V: THERMODYNAMIC POTENTIALS AND PHASE TRANSITIONS

Thermodynamic potentials: Enthalpy- Gibbs and Helmholtz functions- Maxwell relations and their applications-Magnetic work- Magnetic cooling by adiabatic demagnetization-Approach to absolute zero-Change of phase- Equilibrium between a liquid and its vapour- Clausius-clapeyron- Phase transitions-Landau theory of phase transition-Equilibrium between phases-Pomeranchuk cooling.

TEXTBOOKS

- Pathria R.K., Statistical Mechanics, 2nd Edition, Elsevier, 2000
- F. Reif, Fundamentals of Statistical and Thermal Physics, Wave Land Price

References

- K. Huang, Statistical Physics, 2nd edition, John Wiley's and Sons.
- R. K. Pathria, Statistical Mechanics, 2nd edition, Butterworth Heinemann.
- F. Reif, Fundamentals of Statistical and Thermal Physics, Wave Land Price

3.4 Phys 605:Classical Mechanics ((3Cr.Hrs/ 6 ECTS)

INSTRUCTIONAL OBJECTIVES

1. To give students a solid foundation in classical mechanics.

2. To introduce general methods of studying the dynamics of particle systems.

3. To give experience in using mathematical techniques for solving practical problems.

4. To lay the foundations for further studies in physics and engineering.

3.4.1 UNIT I: LAGRANGIAN FORMULATION

Examples of constraints-Principle of virtual work-Lagrange's equations of first kind-D'Alembert's Principle-Lagrangian formulation - Degrees of freedom and generalized coordinates- Euler-Lagrange equations of motions- Simple applications-Linear Harmonic oscillator-Isotropic oscillator- Attwood's machine-Motion of a particle under earth gravity-Simple pendulum-Invariance of Euler-Lagrange equations of motion under generalized coordinate transformations.

3.4.2 UNIT II: HAMILTONIAN FORMULATION

Hamilton's equation of motion from Lagrangian by Legendre's dual transformation- Properties of the Hamiltonian- Configuration space-Phase space-State space-Lagrangian and Hamiltonian of relativistic particles- Hamilton's principle - Derivation of Hamilton's and Euler-Lagrange equations of motion- Invariance of Hamilton's principle under generalized coordinate transformation- Hamilton's principal and characteristic functions.

3.4.3 UNIT III: CANONICAL TRANSFORMATIONS

Definition of canonical transformations- Generating functions- Conditions for canonicality- Poisson brackets-Poisson's theorem- Jacobi-Poisson theorem- Invariance of Poisson brackets under canonical transformation-Hamilton Jacobi equation-Connection with canonical transformation- Applications to simple problems-One dimensional simple harmonic oscillator-Swinging Atwood' s machine-Action-Angle variables.

3.4.4 UNIT IV: RIGID BODY DYNAMICS

Rigid body - Independent co-ordinates of a rigid body-Orthogonal transformation- Eulerian angles- Euler's equation of motion-Angular momentum and kinetic energy of motion about a point- Theorems on moment of inertia and moment of inertia tensors-Eigen values and principal axis transformation-Torque free motion of a rigid body-Heavy symmetrical top with one point fixed-Chandler wobbling of the Earth-Steady precession of a symmetric top under external torque.

3.4.5 UNIT V: SMALL OSCILLATIONS

Types of equilibrium and the potential at equilibrium-Study of small oscillations using generalized coordinates-Eigen value equations-Normal modes of vibrations and normal co-ordinates-Non-degenerate and degenerate systems-Linear Triatomic molecule-Introduction to the Lagrangian and Hamiltonian formulations for continuous systems-Transition from discrete to continuous systems.

TEXTBOOKS

- Rana N. C and Joag P.S., Classical Mechanics, 1st Edition, 2011.
- Herbert Goldstein, Charles P. Poole and John L. Safko, Classical Mechanics, 3rd Edition, Pearson, 2011.

References

- H. Goldstein, Classical Mechanics, Addison Wesley 3rded., 2001
- Marion Thoronton, Classical Dynamics of Particles and Systems, 4thed., 1995
- Murrey R. Speigle, Schaum's Outline series: Theory and problems of theatrical mechanics
- R. Taylor, Calassical Mechanics, Universal Science, 2005
- K. R. Symon, Mechanics, Addison Wesley 3rded., 1971.

3.5 Phys 606: Quantum Mechanics (3Cr.Hrs/ 7 ECTS)

INSTRUCTIONAL OBJECTIVES

- To illustrate the inadequacy of classical theories and the need for a quantum theory.
- To explain the basic principles of quantum mechanics.
- To develop solid and systematic problem solving skills.
- To apply quantum mechanics to simple systems occurring in atomic and solid state physics.

3.5.1 UNIT I: GENERAL FORMALISM

Postulates of quantum mechanics - Wave function and its Physical Interpretation-SchrÃ¶dinger equation-Time-independent SchrÃ¶dinger equation- Dynamical variables and operators- Commutation relations of operators- Hermitian operators- Expansion in Eigen functions- Heisenberg Uncertainty relation- Time evolution operator-SchrÃ¶dinger and Heisenberg pictures of time evolution- Time variation of expectation values.

3.5.2 UNIT II: DISCRETE EIGENVALUE PROBLEMS

Harmonic oscillator in one dimension - Analytic method -Abstract operator method- SchrÃ¶dinger equation in three dimensions - Spherical polar coordinate form- Angular momentum eigen functions and eigen values- Total angular momentum and Spherical harmonics- Radial equation- Hydrogen atom wave functions and their discussion.

3.5.3 UNIT III: PERTURBATION THEORY

Time independent perturbation theory for discrete levels - Non-degenerate cases and degenerate cases-Removal of degeneracy- Spin-Orbit coupling- Fine Structure of Hydrogen- Variational method- Time-dependent perturbation theory - constant and periodic perturbations - Fermi Golden rule - WKB approximation - sudden and adiabatic approximations.

3.5.4 UNIT IV: IDENTICAL PARTICLES AND SCATTERING THEORY

of mass frames- System of identical particles - Symmetric and anti symmetric wave functions- Two electron atoms - Exchange interactions- Spin half particles in a box - Fermi gas- Band structure- Quantum Scattering theory - Differential and total cross sections- Scattering amplitude- Formal expression for scattering amplitude - Green's functions- Born approximation - Application to spherically symmetric potentials.

3.5.5 UNIT V: RELATIVISTIC QUANTUM MECHANICS

The Klein-Gordon (KG) equation - Charged particle in an electromagnetic field, Interpretation of the KG equation- Dirac equation, free particle solution, Negative energy states-Equation of continuity-Plane wave solutions of the Dirac Equation- Non-relativistic limit of the Dirac equation- Fine structure of Hydrogen-Dirac equation in Electromagnetic field.

TEXT BOOK

- J.J. Sakurai, Modern Quantum Mechanics, Addison-Wesley Publishing Company, 1994.
- David J. Griffiths, Introduction to Quantum Mechanics, 2nd Edition, Pearson, 2009.
- Mathews P.M. and Venkatesan K., Quantum Mechanics, 2nd Edition, McGraw Hill, 2010.

References

- Quantum Mechanics L. Schiff
- Quantum Mechanics E. Merzbacher
- Practical Quantum Mechanics S. Flugge
- Quantum Mechanics Mathews and Venkatesan
- Quantum Mechanics M. P. Khanna
- Principles of Quantum Mechanics P. A. M. Dirac

3.6 Phys 608: Computational Physics (3Cr.Hrs/ 6 ECTS)

Numerical solutions of partial differential equation I, Partial Differential equation II, Partial Differential Equation III, Stochastic Methods, Special Functions and quadrature, Numerical Methods: Finding Roots - Newton-Raphson method; Interpolation -Lagrange and cubic spline interpolations, Numerical integration - Trapezoidal and Simpson's rule,The simple pendulum, multidimensional (Monte Carlo) numerical integration; Ordinary differential equation, - Euler methods, Runge-Kutta Methods, second order differential equation, Phase space of a simple harmonic oscillator, One-Dimensional Schrodinger equation (example with anharmonic potential);Partial differential equation - Laplace equation, wave equations and heat equation, Dirichlet, Neumann and Cauchy boundary conditions,Steady state heat equation,Fourier analysis: Ranging applications (sonar and radar) (Students must be able to write code themselves in each topic with Fortran/C or Matlab to develop mathematica module)

1. Numerical solutions of partial differential equation I, Partial Differential equation II, Partial Differential Equation III, Stochastic Methods, Special Functions and quadrature,
2. Numerical Methods: Finding Roots ;Newton-Raphson method; Interpolation; Lagrange and cubic spline interpolations, 2.3 Numerical integration ; Trapezoidal and Simpson's rule,The simple pendulum, multidimensional (Monte Carlo) numerical integration; Ordinary differential equation, - Euler methods, Runge-Kutta Methods, second order differential equation, Phase space of a simple harmonic oscillator, One-Dimensional Schroedinger equation (example with anharmonic potential); Partial differential equation; Laplaces equation, wave equations and heat equation (Students must be able to write code themselves in each topic with Fortran/C OR develop mathematica module or Matlab)

Refrences

- Franklin J. - Computational Methods for Physics Cambridge (2013)Press,
- De Vries P. L. - A first course in computational physics , John wiley & Sons, New York (1994) New York (1994)

- S. C. Chapra and R. P. Canale, Numerical Methods for Engineers with Software and Programming Applications, Fourth Edition, McGraw-Hill, 2002.
- H. Gould, J. Tobochnik, W. Christian, An Introduction to Computer Simulation Methods, Third Edition, Pearson Education, 2007.
- Anagnostopoulos K .N. - Computational Physics , National Technical University, (2014)Athens
- Koonin S. E., Meredith D. C. - Computational Physics Computational Westview press (1990).
- S. J. Chapman, Fortran 95/2003 for Scientists and Engineers, Third Edition, McGraw-Hill, 2007
- S. E. Koonin and D. C. Meredith, Computational Physics, Fortran Version, Westview Press, 1990

3.7 Phys 711: Quantum Optics I (3Cr.Hrs/ 7 ECTS)

I. **Aim**

The aim of this course is to enable students to acquire a working knowledge of certain quantum states of light and some quantum distribution functions. II. **Content**

The quantum radiation field, Number states, Coherent states, Chaotic states, Squeezed states, The P function, The Q function, The Wigner function, The photon number distribution, The photon count distribution, The master equation for a cavity mode, The Fokker-Planck equation, The quantum Langevin equation, Input-output relation

III.**Learning Outcomes**

At the end of this course, the students will be familiar with various quantum distribution functions and equations of evolution of cavity modes.

1. Fesseha Kassahun, Refined Quantum Analysis of Light (CreateSpace Independent Publishing Platform, 2014).
2. M.O. Scully and M.S. Zubairy, Quantum Optics (Cambridge University Press, Cambridge, 1997).
3. D.F. Walls and G.J. Milburn, Quantum Optics (Springer-Verlag, Berlin, 1994).

3.8 Phys 713 - Quantum Optics II (7 ECTS)

I. **Aim**

The aim of this course is to enable students to acquire a working knowledge of atom-radiation interaction, resonance fluorescence, and atom cooling. II. **Content**

The atom-radiation interaction Hamiltonian, Atomic dynamics, Collapse and revival, The master equation for a two-level atom, Resonance fluorescence: Photon antibunching, Photon statistics, Power spectrum, Cooling of a two-level atom: The retarding force, The effective temperature III. **Learning Outcomes**

At the end of this course, the students will be familiar with the quantum properties of the fluorescent light from a two-level atom, and the cooling of a two-level atom by coherent light. Iv. **References**

1. At the end of this course, the students will be familiar with the quantum properties of the fluorescent light from a two-level atom, and the cooling of a two-level atom by coherent light.

2. M.O. Scully and M.S. Zubairy, Quantum Optics (Cambridge University Press, Cambridge,1997).

3. D.F. Walls and G.J. Milburn, Quantum Optics (Springer-Verlag, Berlin, 1994).

3.9 Phys 721:Solid State Physics I(3Cr.Hrs/ 7 ECTS)

3.9.1 Content

- **Crystal Structure**: Bravais lattices, crystal systems -point groups, space groups and typical structures, Reciprocal Lattice, Planes and directions -Point, line, surface and volume defects

- **Ionic crystals**: Born Mayer potential, Thermochemical Born-Haber cycle

- **Van der Waals binding**: rare gas crystals and binding energies-Covalent and metallic binding: characteristic features and examples Crystal

- **Diffraction and Lattice Vibrations**: X-rays, neutrons, electrons -Bragg's law in direct and reciprocal lattice -Structure factor -diffraction techniques

- **Lattice Dynamics**: monatomic and diatomic lattices, Born-von Karman method, Phonon frequencies and density of states, Dispersion curves, inelastic neutron scattering
- **Thermal and Elastic Properties**: Reststrahlen Specific heat, Thermal expansion, Thermal conductivity, Normal and Umklapp processes, Propagation of elastic waves and measurement of elastic constants
- **Conductors and Semiconductors**: Free electron theory of metals;Thermal and transport properties; Bloch functions; Nearly free electron approximation; Formation of energy bands. Kronig Penny Model; Brillouin zone; Effective mass; concept of holes, Fermi surface ; Semiconductors: carrier statistics in intrinsic and extrinsic crystals; electrical conductivity; Hall effect Electronic specific heat.
- **Super Conducting**: Optical and Dielectric materials Superconductors: Properties, BCS theory, Flux quantization, Josephson effects, High Tc superconductors, Applications
- **Optical Materials**: Optical absorption, colour centres, Trap, recombination, excitons, Photoconductivity, luminescence
- **Dielectrics:** Macroscopic electric field, Local electric field in an atom, dielectric constant and polarizability, Clausius - Mossotti equation, measurement of dielectric constant, Ferroelectrics
- **Magnetic Materials**: Magnetic materials: Types, Quantum theories of dia and para magnetism
- **Susceptibility Measurement**: Guoy Balance, Quincke's method; Ferromagnetic order, Hysterisis, Curie point and exchange integral, Magnons, domain theory ; Ferri and antiferrimagnetic order, Curie temperature, susceptibility and Neel Temperature.

References

- Introduction to Solid State Physics by Kittel.
- Patterson: Solid State Physics
- R. A.Levy : Principles of Solid State Physics
- Mckelvy: Solid State and Semi-conductor Physics.

- J. S. Blakemore Solid State Physics
- J. Ziman Principles of the Theory of Solids

3.10 Phys 723:Solid State Physics II (3Cr.Hrs/ 7 ECTS)

Content

Surface effects, Thermionic emission, Schottky emission, Field emission, Classical and quantum theory of harmonic crystals, Anharmonic effects in crystals, Thermal expansion and Gruneisen parameter, Phonon dispersion relation in metals, Quantum well structures, Semiconductor lasers, Diamagnetism, Paramagnetism, Magnetic ordering, Magnetic structures, Ferro-magnetism, Ferri-magnetism, Magnons, Spin waves, Superconductivity, Meissner effect, Energy gap, London equation, BCS theory of superconductivity, Flux quantization in a superconducting ring, Josephson effect, Point defects and dislocations.

References

- Introduction to Solid State Physics by Kittel.
- Patterson: Solid State Physics
- R. A.Levy : Principles of Solid State Physics
- Mckelvy: Solid State and Semi-conductor Physics.
- J. S. Blakemore Solid State Physics
- J. Ziman Principles of the Theory of Solids

3.11 Phys 731 Seminar(1Cr.Hrs/ 7 ECTS)

Content

Approaches in scientific writing, elements of scientific paper, seminar writing and presentation based on sound scientific paper.

3.12 Phys 602: Research Method in physics(1Cr.Hrs/ 7 ECTS)

Aim and Scope This is a cross-curricular subject, which may be of interest for those students who are considering undertaking a research career, especially in the fields of physics and technologies in physics. The goal of the course is to introduce students to the research methodology in science. To do so, both theoretical and applied aspects will be combined in the four parts the course is divided in:

- The process of scientific research
- Ethical aspects of the research work
- Introduction to scientific policies.
- Communication techniques.

I. **Unit 1: The process of scientific research.**
Research methodology and research methods, methodology -selecting a topic, hypothesis, experiment, analysis and results; non-experimental or theoretical research, critical thinking, investigation, survey, ab initio, semi-empirical, empirical search; Inquiry, Quest, Exploration, innovation (innovative ideas), Discovery and Invention in science; knowledge and creativity A flow-chart of research leading to degree, Literature survey through web and other related resources, scientific report/paper writing, documentation in Latex,The scientific method, research planning; the scientific explanation and demarcation,criteria,

characteristics of factual sciences, scientific epistemology, technology as transformational knowledge, relations between science and technology, the researcher and the structure of the research teams, Quality of research, quantitative measurement by Impact factor, h-index, Scientometry,

II. **Unit 2:Mathematical Techniques**

Uncertainties in Measurements: Measuring Errors, Uncertainties, Parent and Sample Distributions, Mean and Standard Deviation of Distributions, Binomial Distributions, Poisson Distribution, Gaussian or Normal Error Distribution, Lorentzian Distribution; Approximation and Errors in Computing: Significant Digits, Numerical Errors, Modelling errors, Conditioning and Stability, Convergence of Iterative Processes. Error Analysis: Instrumental and Statistical Uncertainties, Propagation of Errors, Application of Error Equations, method of Least squares, Statistical Fluctuations, Probability Tests, χ^2 Test of a distribution,Curve fitting (Regression Analysis); Least square Fit to a Straight line, error estimation of the fitted parameters, limitations of the least square method, Least squares fit to a polynomial, matrix solution, Goodness of a fit, Linear Correlation Coefficient, Multivariable Correlations, Monte Carlo techniques: Random numbers, Random numbers from Probability Distributions, Specific Distributions, Efficient Monte Carlo generation, Confidence Intervals, Monte Carlo Tests for the Fit- Applications. Numerical solutions of Transcendental Equations and Ordinary Differential Equations, Runge-Kutta Methods III. **Unit 3: Ethical aspects of the research work, scientific policies and Ethical aspects of the research work**

Scientific ethics, axiology and ethical values of science, ethics of the researcher, personal code of conduct, internal code of conduct, conduct guidelines, ethical standards of publication, scientific fraud and malpractice; study of historical and contemporary cases,Typology of research projects, strategic plans and guidelines, research products: open access publications, patents, utility models, trade secret, etc.; training of researchers, preparation of research projects, monitoring and evaluation processes,Scientific ethics, axiology and ethical values of science, ethics of the researcher, personal code of conduct, internal code of conduct, conduct guidelines, ethical standards of publication, scientific fraud and malpractice; study of historical and contemporary cases IV. **Unit 4: Communication techniques and Scientific Policies.**

Dissemination of results, technical and scientific documents, characteristics and quality indices of journals, structure of scientific documents, preparation of written documents (research articles, reports), techniques of oral presentation and defense of research works, other formats (posters, flash presentations, etc.), skills for academic writing , online communication technologies, evaluation procedures,Typology of research projects, strategic plans and guidelines, research products: open access publications, patents, utility models, trade secret, etc.; training of researchers, preparation of research projects, monitoring and evaluation processes.

References

- Marder, M. P. (2011). Research methods for science. Cambridge University Press.
- Paler-Calmorin, Laurentina. Methods of Research and Thesis Writing, 2006. .
- Rex Bookstore, Inc. Manila, Philippines Temechegn Engida. Educational Research Methods (Module), 2008.
- Louis Cohen, Lawrence Manion and Keith Morrison. Research Methods in Education
- Joseph Gibaldi. MLA Handbook for Writers of Research Paper 6th ed.,. First East-West Press Edition, New Delhi, 2004
- N. M. Glazunov FOUNDATIONS OF SCIENTIFIC RESEARCH,National Aviation University,2012

3.13 Phys 732:Selected Topics (3Cr.Hrs/ 7 ECTS)

Content

A course in which various selected topics in physics are covered based on the availability of current topics related to the specialization of the graduate.

3.14 Phys 742: Graduate Thesis (6 Cr.Hrs/ 28 ECTS)

Content

A research projects (Theoretical, Experimental or Computational) to be carried out in different areas of physics under the supervision of advisor.